

Problem Statement
A commuter rushing to find parking before class, stressed students wasting time finding a place to study, campus visitors searching through lines of parked cars for a spot on gameday- what is the common theme? Rutgers, amongst other state schools, has a burgeoning student population. This growth is causing congestion within the infrastructure, a lower-quality student experience, and ultimately leads to an all-to-familiar frustration. Other areas in which students have reported issues are overcrowded buses, full dining halls, and long lines to enter stadiums on gameday. Scarlet Impact set out to resolve this intimidating task, working to create a scalable, all-encompassing solution that will touch the lives of all who call the university home.

Target Audience/Beneficiaries
UniView differentiates itself through its benefits to target key demographics. An often overlooked group, commuters will benefit immensely from the open spot parking sensor, however, this will serve any student traveling between campuses. The high occupancy of the libraries, recreation centers, dining halls, and academic buildings reflect a fraction of the users that would benefit from this capacity-sensing technology. Faculty will also benefit from UniView when finding parking in crowded faculty lots. Finally, our solution serves a crucial demographic, campus visitors, in search of parking, particularly on Rutgers game days.

Proposed Solution
Using commercially available IoT cameras, our solution can stitch together various camera frames at different angles to deliver an accurate picture of activity at the locations outlined above. Utilizing Verizon's low-latency, 5G networks in combination with AWS servers, UniView can seamlessly and reliably transfer data from our collection points to the end user, whether it be Rutgers students or administration seeking to optimize the infrastructure. UniView's parking availability feature is an opportunity to make use of the rising spread of solar panel technology to power the sensors, which are already present above most parking lots within Rutgers University. By positioning cameras with fishbowl lenses above pedestrians in parking lots and within facilities the cameras can collect nonpersonal data, in addition to anonymizing the data downstream. Thus, UniView can gather data points while maintaining the integrity of private data collection. Additionally, by deriving a holistic view of the points of interest we can provide real-time analytics to Rutgers students, delivering relevant insights on available parking, low-traffic dining halls, and available study spots through dashboards on the mobile app. This eliminates time wasted traveling between campuses to only find out

that there is no available seating to study, or driving up and down parking lots inconveniently looking for an available parking spot. Another crucial feature is the ability to store data and derive insights. For example, examining crowds on buses at various times could help optimize public transportation, helping determine the number of vehicles that need to be in service at any given time. By integrating Verizon's 5G networks, Multi-Access Edge Computing, IoT devices, and AWS cloud software, UniView can dramatically improve the Rutgers experience, making this scalable solution beneficial for the students, faculty, and university.

Impact and Value Proposition

UniView, targeted mainly towards the students and faculty, would also benefit visitors coming to Rutgers. It would provide them with popular student locations and allow them to view available parking spots. Uniview poses a solution that would reduce the congestion of Rutgers; it is also an app that can make a profit for Rutgers or Verizon while benefiting the students. The app's revenue model would be based on a combination of user fees for premium features, such as guaranteed parking spots and priority booking of locations. With the sensors used for parking, UniView can allow the user to view available parking spots in different places. As users are looking for parking in the lots they are permitted to park in, another option would be to park in faculty or other lots while paying a fee less than a parking ticket. It would allow the user to see the spot, and based on the location, the parking spot can be available for purchase, which would be lower than if the user were to park illegally in that specific lot. This portion of the app would be highly beneficial for the users as they are not bound to park in a particular location, giving them access to view what is available. In hopes of solving the overcrowding issue at Rutgers, UniView would provide users with access and premium features that would enhance everyone's experience at any campus at Rutgers.

Innovation and Feasibility

Privacy regulations are of the utmost importance when discussing implementing complex digital solutions. Legal constraints relating to privacy include the New Jersey Identity Theft Prevention Act and the New Jersey Data Privacy Act, which restrict the collection of personally identifiable information. To ensure compliance, UniView implements encryption, data anonymization, and participates in timely deletion to follow privacy best practices. Financial considerations were also considered, leading to the selection of commercially available cameras, significantly reducing the cost of procurement and installation. Lastly, we ensured scalability by using IoT devices and software that can continually collect and transfer large amounts of data without large investments into the maintenance of the solution's physical and cloud infrastructure.

Conclusion

Uniview is simple yet effective, solving issues experienced by the entire Rutgers community. Verizon's 5G technology which enables real-time data collection, transmission, and processing can communicate directly with the user's app making this project an impactful, scalable application of Verizon's low-latency 5G network across universities. In this manner, UniView's subtle IoT sensors have the potential for large-scale impact.